

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

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COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA5604

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications.

Aim

To design and simulate a **Multi-access Edge Computing (MEC)** architecture for smart traffic applications and analyze how edge processing reduces communication latency compared with cloud-only processing.

Objective

- To collect traffic data from connected sensors and cameras.
- To process time-critical traffic information at the edge.
- To reduce the delay caused by sending all data to a remote cloud.
- To provide faster traffic-control decisions.
- To demonstrate latency improvement using MATLAB.

Program

clc;

clear;

close all;

% MEC and Cloud latency (ms)

MEC = 5 + 8 + 5;

Cloud = 5 + 40 + 25 + 5;

% Latency reduction

Reduction = ((Cloud - MEC)/Cloud)*100;

% Display results

```
fprintf('MEC Latency = %.2f ms\n', MEC);
```

```
fprintf('Cloud Latency = %.2f ms\n', Cloud);
```

```
fprintf('Latency Reduction = %.2f %%\n', Reduction);
```

% Plot comparison

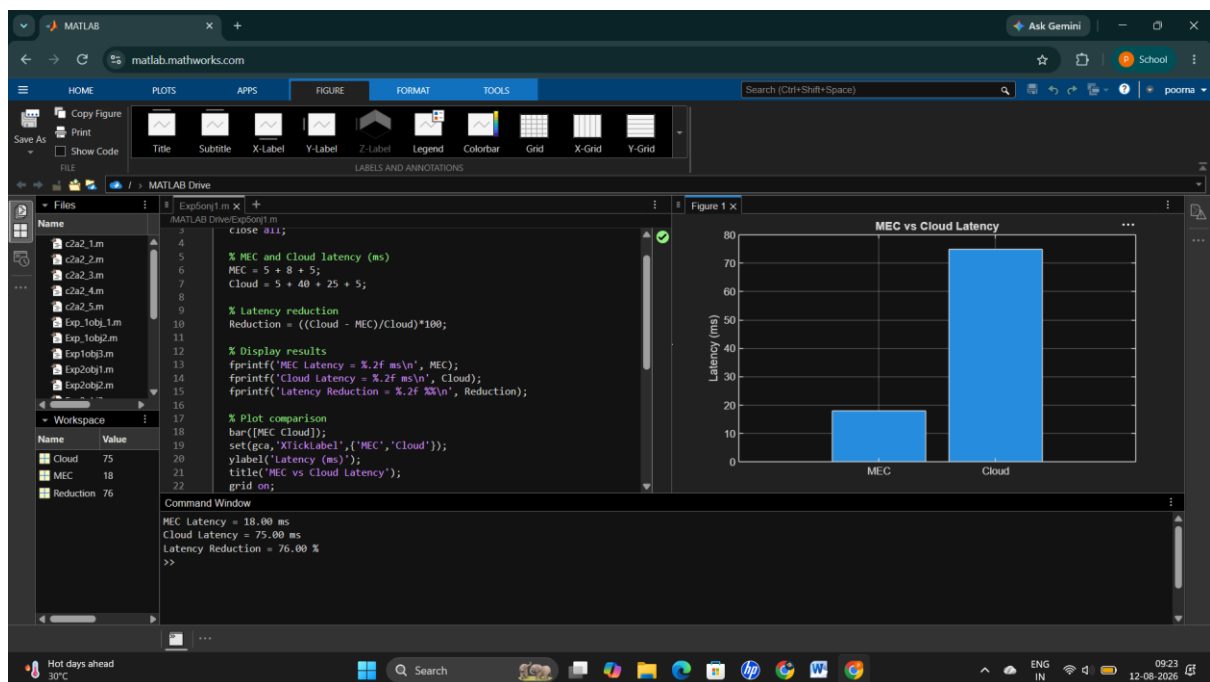
```
bar([MEC Cloud]);
```

```
set(gca,'XTickLabel',{'MEC','Cloud'});
```

```
ylabel('Latency (ms)');
```

```
title('MEC vs Cloud Latency');
```

```
grid on;
```



Result

Thus, a **MEC-based smart traffic architecture** was designed and simulated successfully. The simulation demonstrates that processing traffic information at the edge can significantly reduce latency and enable faster traffic-control decisions.

Title

2. Design of a Satellite-Assisted Communication Model Using LEO Satellite for Remote Wireless Coverage

Aim

To design and simulate a **Low Earth Orbit (LEO) satellite-assisted communication system** for providing wireless coverage to remote and underserved areas.

Objective

- To design a communication architecture using LEO satellites.
- To connect remote users to the terrestrial network.
- To calculate free-space path loss.
- To calculate received signal power.
- To analyze the effect of LEO satellite altitude on communication.
- To simulate the link using MATLAB.

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
c = 3e8;
```

```
f = 20e9;
```

```
lambda = c/f;
```

```
Pt = 40;
```

```
Gt = 30;
```

```
Gr = 20;
```

```
d = 500:500:2000;
```

```
FSPL = 20*log10((4*pi*d*1000)/lambda);
```

```
Pr = Pt + Gt + Gr - FSPL;
```

```
fprintf('Wavelength = %.4f m\n',lambda);

fprintf('Distance(km) FSPL(dB) Received Power(dBm)\n');
```

```
for i = 1:length(d)

    fprintf('%5d    %.2f    %.2f\n',d(i),FSPL(i),Pr(i));

end
```

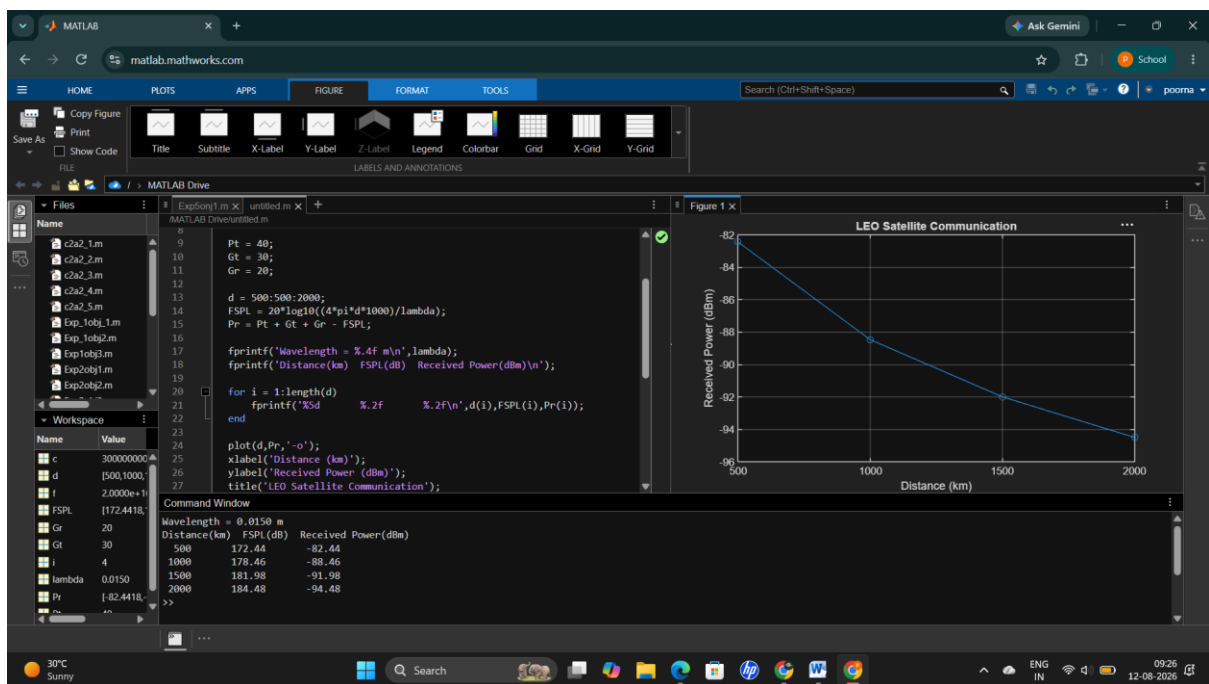
```
plot(d,Pr,'-o');

xlabel('Distance (km)');

ylabel('Received Power (dBm)');

title('LEO Satellite Communication');

grid on;
```



Result

Thus, a **LEO satellite-assisted communication architecture** was designed and simulated successfully. The MATLAB simulation demonstrates the relationship between satellite distance, free-space path loss, and received signal power, showing the suitability of LEO satellites for providing wireless connectivity to remote areas.